

Manaswini Ganjam

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PROFESSIONAL SUMMARY

Highly motivated to integrate modeling tools and geospatial skills for effective environmental management, with a focus on contributing to the research and science of climate change mitigation.

EDUCATION

North Carolina State University

Ph.D. in Forestry and Environmental Resources

Raleigh, NC

Expected May 2025

Jawaharlal Nehru Technological University

M. Tech in Environmental Geomatics

Hyderabad, India

August 2015

B. Tech in Electrical and Electronics Engineering

May 2013

TECHNICAL SKILLS

- R
- PostgreSQL
- Arc GIS/ QGIS
- Python
- GEE / AWS (Geospatial cloud data)
- C, C++

EXPERIENCE

North Carolina State University

Research Assistant

Raleigh, NC

May 2021 – Expected May 2025

- Conducted high-level research on carbon sequestration across North American biomes, focusing on climate change impacts on forest productivity.
- Implemented and calibrated a process model (Biome BGC MuSo) to simulate carbon fluxes, enhancing understanding dynamics of net ecosystem productivity with stand succession.
- Developed geospatial database by processing diverse datasets (FIA, NLCD, gSSURGO, meteorological data) for the southeastern United States, utilizing statistical methods to ensure data accuracy and reliability.
- Generated visualizations and interpreted model outputs to communicate complex scientific concepts of carbon flux dynamics, applying statistical inferences to support data-driven conclusions.
- Contributed to multiple research publications and presentations, showcasing ability to share research outcomes with the scientific community.
- Conducted comprehensive literature review on ecological forest carbon models, broadening expertise in various quantitative assessment methods for carbon flux estimation.
- Advanced programming skills (Python, R, PostgreSQL) for large-scale geospatial data analysis and process model input preparation.
- Collaborated with forest economists to understand carbon credit impacts on sequestration, informing management decisions and climate mitigation strategies.

Central Coffee Research Institute

Project Scientist

Bengaluru, India

June 2017 – March 2018

- Performed pre-processing of high spatial resolution remote sensing data (1m).
- Conducted multiple fieldwork trips for GPS-based ground truth collection.
- Utilized object-based classification for the classification and interpretation of remote sensing data.
- Contributed to the generation of a coffee area map for India.

CSIR-National Geophysical Research Institute

Project Assistant III

Hyderabad, India

November 2015 – March 2017

- Requisitioned satellite datasets like land use, elevation, and vegetation density.
- Digitized high resolution bathymetry data.
- Conducted preprocessing of data, database creation, ancillary data collection.

- Inundation results were obtained using a numerical tsunami model.
- Conducted literature review to assign weights to the parameters using analytical hierarchy process.
- Generated vulnerability maps for Car Nicobar Islands in Indian ocean.

National Remote Sensing Centre

Hyderabad, India

Project intern

September 2014 – October 2015

- Acquired and managed multi-temporal remote sensing datasets, including Landsat and AWIFS imagery.
- Performed comprehensive data preprocessing, database creation, and ancillary data collection for geospatial analysis
- Implemented unsupervised classification techniques to assess geographical areas affected by forest fires, deforestation, and degradation.
- Conducted in-depth geospatial and statistical analyses to derive meaningful insights from remote sensing data.
- Contributed to the writing and publication of research articles, while actively collaborating on related research projects resulting in multiple peer-reviewed publications.

PUBLICATIONS

- **Ganjam, M.**, Baker, J.S., Aguilos, M., Merganićová, K. (2024). Maximizing Net Ecosystem Productivity: Integrating Chronosequence Analysis and Ecosystem Modeling Across Contrasting Biomes in North America. Forests (to be submitted).
- Fuller, M. R., **Ganjam, M.**, Baker, J. S., & Abt, R. C. (2024). Advancing forest carbon projections requires improved convergence between ecological and economic models (Under review).
- Reddy, C.S., Jha, C.S., **Manaswini, G.**, Alekhya, V.V.L.P., Vazeed Pasha, S., Satish, K.V., Diwakar, P. G., & Dadhwal V.K. "Nationwide assessment of forest burnt area in India using Resourcesat-2 AWiFS data". Curr. Sci (2017).
- Reddy, C.S., **Manaswini, G.**, Jha, C.S., Diwakar, P.G., & Dadhwal, V.K. "Development of national database on long-term deforestation in Sri Lanka". Journal of the Indian Society of Remote Sensing, 1-12 (2016).
- Reddy, C.S., **Manaswini, G.**, Satish, K.V., Sonali Singh., Jha, C.S., Dadhwal V.K. "Conservation priorities of forest ecosystems: Evaluation of deforestation and degradation hotspots using geospatial techniques". Ecological Engineering volume 91 (2016):333–342.
- **Manaswini, G.**, and Reddy, C.S., "Geospatial monitoring and prioritization of forest fire incidences in Andhra Pradesh, India". Environmental monitoring and assessment 187.10 (2015):1-12.
- Jakkula, M., Guruhappa, H., **Manaswini, G.**, & Srivastava, K. "Tsunami forces acting on ocean structures: A synthetic study". Journal of Indian Geophysical Union, 21(6), 482-489 (2017).